When Verbs Forget: Evidence for Negatively Accelerated Decay in Sentence Processing

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Linguistic representations in working memory decay over time, but the temporal dynamics of this decay in sentence processing remain unclear. Prominent sentence processing theories such as Dependency Locality Theory (Gibson 2000) and Lewis and Vasishth's (2005) influential ACT-R model assume non-linear decay, yet there is no direct linguistic evidence supporting this assumption.

We investigated this question in two English acceptability-judgment experiments involving coordinate structures. Both experiments manipulated the distance between a verb and a complement that it does not typically select for, following the design logic of Kim and Lu (2024) while removing several confounds in their materials (not discussed here due to space constraints).

Experiment 1 tested the acceptability of items such as (1), where the verb, *depend*, combined with a coordination of three conjuncts: two PPs headed by *on* and one CP. The critical manipulation varied the CP's position within the coordination: first (CP1; (1a)), second (CP2; (1b)), or third (CP3; (1c)).

(1) a. The project **depends** [[$_{\text{CP}}$ that the new outsourced contractors will do their part], [$_{\text{PP}}$ on an appropriate engineering design], and [$_{\text{PP}}$ on the diligence of everybody involved]].
b. The project **depends** [[$_{\text{PP}}$ on an appropriate engineering design], [$_{\text{CP}}$ that the new outsourced contractors will do their part], and [$_{\text{PP}}$ on the diligence of everybody involved]].
c. The project **depends** [[$_{\text{PP}}$ on an appropriate engineering design], [$_{\text{PP}}$ on the diligence of everybody involved], and [$_{\text{CP}}$ that the new outsourced contractors will do their part]].

Because *depend* is a verb that resists declarative CP complements, we predicted that placing the CP later would increase acceptability: later integration of the CP causes the verb's selectional requirements to decay more ("structural forgetting"; Gibson and Thomas 1999), making the violation less noticeable ("grammaticality illusion"; Phillips *et al.* 2011).

In a norming study, in addition to *depend*, we identified three more verbs that readily take PPs but resist declarative CP complements: *speak*, *suffer*, *talk*. For each of the four verbs, we constructed six items patterned on (1), with all conjuncts matched exactly at 12 syllables. This produced 72 experimental stimuli distributed across three counterbalanced lists.

72 native speakers of English (balanced across UK and US) rated these stimuli using the Thermometer scale (Featherston 2008, 2009). As shown in Figure 1a, acceptability increased monotonically with the distance of the CP from the verb: CP1 $<$ CP2 $<$ CP3. Mixed-effects modeling confirmed significant differences between CP1 and the two other conditions ($p < .001$), while the CP2–CP3 contrast did not reach significance ($p = .12$), failing to support our full prediction. A Likert-scale replication yielded the same pattern, showing that results were not method-dependent.

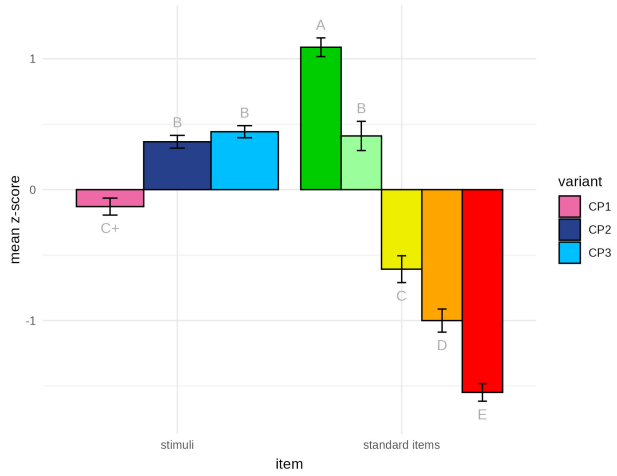
We subsequently hypothesized that these results might reflect the non-linear, negatively accelerated decay of linguistic representations: with longer conjuncts in Experiment 1, substantial forgetting may have already occurred during the first conjunct, leaving little additional decay for later positions.

To test this, Experiment 2 used the same design but halved the length of each conjunct to six syllables (see (2)) to make position-dependent differences more salient under the non-linearity assumption.

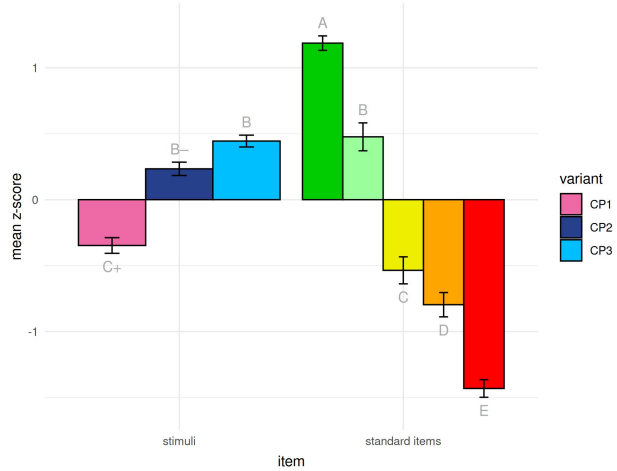
(2) The project **depends** [[$_{\text{CP}}$ that the morale stays high], [$_{\text{PP}}$ on stakeholder input], and [$_{\text{PP}}$ on timely decisions]].

As shown in Figure 1b, judgments from 77 native speakers of English revealed the same monotonic pattern as in Experiment 1 (CP1 $<$ CP2 $<$ CP3), but, in Experiment 2, mixed-effects modeling detected significant differences at all three distances (all $p < .001$). Taken together with the results of Experiment 1, these findings support a negatively accelerated decay function where the initial decay is steep but plateaus after roughly 20 syllables.

In sum, these experiments provide direct evidence that linguistic representations in working memory decay in a non-linear, negatively accelerated fashion. Notably, the initial steep decay plateaus rather early, after which further forgetting is minimal. Interestingly, our findings further show that such distance-based decay is sufficient to create grammaticality illusions, independent of interference effects invoked for other types of illusions documented in the literature.



(a) Mean z-scores of stimuli and standard items in Experiment 1 (with 95% *CI*s)



(b) Mean z-scores of stimuli and standard items in Experiment 2 (with 95% *CI*s)

Figure 1: Results of Experiments 1 and 2 (about standard items occurring there, see Featherston 2008, 2020; in brief, A–B are grammatical, C is borderline, D–E are ungrammatical.)

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